Experiment: 7

```
/* Write a program to illustrate recursive function */

#include <stdio.h>

int sum(int k)
{
  if (k > 0)
  {
    return k + sum(k - 1);
  }

}

int main()
{
  int a;
  printf("\n Enter a positive integer number: ");
  scanf("%d",&a);
  if(a>0)
  {
  int result=sum(a);
  printf("\n Sum from 1 to %d is: %d",a,result);
  }
  else
  {
    printf("\n Enter valid input");
  }
  return 0;
}
```

Output:

Enter a positive integer number: 17

Sum from 1 to 17 is: 153

Experiment: 8

```
/* Write a program of structure */

#include<stdio.h>

struct student
{
    int roll;
    char name[50],section[10];

};

int main()
{
    struct student s1,s2;

    // Enter first student's details
    printf("\n Enter first student's roll number: ");
    scanf("%d",&s1.roll);
    printf("\n Enter first student's name: ");
    scanf("%s",s1.name);
    printf("\n Enter first student's section: ");
    scanf("%s",s1.section);

    // Enter second student's details
    printf("\n Enter second student's roll number: ");
```

```
    scanf("%d",&s2.roll);
    printf("\n Enter second student's name: ");
    scanf("%s",s2.name);
    printf("\n Enter second student's section: ");
    scanf("%s",s2.section);

    //print first student's details
    printf("\n \n First student's details: ");
    printf("\n Student's Roll Number: %d",s1.roll);
    printf("\n Student's Name: %s",s1.name);
    printf("\n Student's Section: %s",s1.section);

    //print second student's details
    printf("\n \n Second student's details: ");
    printf("\n Student's Roll Number: %d",s2.roll);
    printf("\n Student's Name: %s",s2.name);
    printf("\n Student's Section: %s",s2.section);

    return 0;
}
```

Output:

Enter first student's roll number: 1001

Enter first student's name: ankit

Enter first student's section: A

Enter second student's roll number: 1002

Enter second student's name: amit

Enter second student's section: B

First student's details:

Student's Roll Number: 1001

Student's Name: ankit

Student's Section: A

Second student's details:

Student's Roll Number: 1002

Student's Name: amit

Student's Section: B